



Problem 1 based on Arrays

❖ Write a C program to count bits set in an integer?
[2]

❖ What is count bits set (number of 1 bits in binary representation of a 32 bit number) :

Suppose integer is 5,

Binary representation is 00000000000000000000000000000000**101**.

Count bits set = 2.

Suppose integer is 10,

Binary representation is 00000000000000000000000000000000**1010**.

Count bits set = 2.



Problem 1 based on Arrays

- ❖ Write a C program to count bits set in an integer [2]?
 - ❖ There are a number of ways to count the number of bits set in an integer. Here are some C programs to do the same.

❖ Method1:

```
#include<stdio.h>
int main()
{
    unsigned int num=10;
    int ctr=0;

    for(;num!=0;num>>=1)
    {
        if (num&1)
        {
            ctr++;
        }
    }

    printf("\n Number of bits set in [%d] = [%d]\n", num, ctr);
    return(0);
}
```



Problems Based on Arrays

❖ Write a C program to count bits set in an integer [2]?

❖ **Method 2:** This is a faster way of doing the same thing. Here the control goes into the while loop only as many times as the number of bits set to 1 in the integer!.

```
#include<stdio.h>
int main()
{
    int num=10;
    int ctr=0;

    while (num)
    {
        ctr++;
        num = num & (num - 1); // This clears the least significant bit set.
    }

    printf("\n Number of bits set in [%d] = [%d]\n", num, ctr);
    return(0);
}
```



Problems Based on Arrays

❖ Write a C program to count bits set in an integer [2]?

❖ Method 3:

❖ This method is very popular because it uses a lookup table.

❖ This speeds up the computation. What it does is it keeps a table which hardcodes the number of bits set in each integer from 0 to 255.

```
0 - 0 Bit(s) set.  
1 - 1 Bit(s) set.  
2 - 1 Bit(s) set.  
3 - 2 Bit(s) set.  
...
```

Example: Integer Number: 250

- Binary Representation of 32 bit number 250:

00000000 00000000 00000000 11111010



Problems Based on Arrays

❖ Write a C program to count bits set in an integer [2]?

```
const unsigned char LookupTable[] = { 0, 1, 1, 2, 1, 2, 2, 3, 1, 2, 2, 3, 2, 3, 3, 4,
    1, 2, 2, 3, 2, 3, 3, 4, 2, 3, 3, 4, 3, 4, 4, 5,
    1, 2, 2, 3, 2, 3, 3, 4, 2, 3, 3, 4, 3, 4, 4, 5,
    2, 3, 3, 4, 3, 4, 4, 5, 3, 4, 4, 5, 4, 5, 5, 6,
    1, 2, 2, 3, 2, 3, 3, 4, 2, 3, 3, 4, 3, 4, 4, 5,
    2, 3, 3, 4, 3, 4, 4, 5, 3, 4, 4, 5, 4, 5, 5, 6,
    2, 3, 3, 4, 3, 4, 4, 5, 3, 4, 4, 5, 4, 5, 5, 6,
    3, 4, 4, 5, 4, 5, 5, 6, 4, 5, 5, 6, 5, 6, 6, 7,
    1, 2, 2, 3, 2, 3, 3, 4, 2, 3, 3, 4, 3, 4, 4, 5,
    2, 3, 3, 4, 3, 4, 4, 5, 3, 4, 4, 5, 4, 5, 5, 6,
    2, 3, 3, 4, 3, 4, 4, 5, 3, 4, 4, 5, 4, 5, 5, 6,
    3, 4, 4, 5, 4, 5, 5, 6, 4, 5, 5, 6, 5, 6, 6, 7,
    2, 3, 3, 4, 3, 4, 4, 5, 3, 4, 4, 5, 4, 5, 5, 6,
    3, 4, 4, 5, 4, 5, 5, 6, 4, 5, 5, 6, 5, 6, 6, 7,
    3, 4, 4, 5, 4, 5, 5, 6, 4, 5, 5, 6, 5, 6, 6, 7,
    4, 5, 5, 6, 5, 6, 6, 7, 5, 6, 6, 7, 6, 7, 7, 8};
```

```
unsigned int num;
unsigned int ctr;    // Number of bits set.

ctr = LookupTable[num & 0xff] +
    LookupTable[(num >> 8) & 0xff] +
    LookupTable[(num >> 16) & 0xff] +
    LookupTable[(num >> 24)];
```

Problems Based on Arrays

❖ Write a C program to count bits set in an integer [2]?

```
const unsigned char LookupTable[] = { 0, 1, 1, 2, 1, 2, 2, 3, 1, 2, 2, 3, 2, 3, 3, 4,
1, 2, 2, 3, 2, 3, 3, 4, 2, 3, 3, 4, 3, 4, 4, 5,
1, 2, 2, 3, 2, 3, 3, 4, 2, 3, 3, 4, 3, 4, 4, 5,
2, 3, 3, 4, 3, 4, 4, 5, 3, 4, 4, 5, 4, 5, 5, 6,
1, 2, 2, 3, 2, 3, 3, 4, 2, 3, 3, 4, 3, 4, 4, 5,
2, 3, 3, 4, 3, 4, 4, 5, 3, 4, 4, 5, 4, 5, 5, 6,
2, 3, 3, 4, 3, 4, 4, 5, 3, 4, 4, 5, 4, 5, 5, 6,
3, 4, 4, 5, 4, 5, 5, 6, 4, 5, 5, 6, 5, 6, 6, 7,
1, 2, 2, 3, 2, 3, 3, 4, 2, 3, 3, 4, 3, 4, 4, 5,
2, 3, 3, 4, 3, 4, 4, 5, 3, 4, 4, 5, 4, 5, 5, 6,
2, 3, 3, 4, 3, 4, 4, 5, 3, 4, 4, 5, 4, 5, 5, 6,
3, 4, 4, 5, 4, 5, 5, 6, 4, 5, 5, 6, 5, 6, 6, 7,
2, 3, 3, 4, 3, 4, 4, 5, 3, 4, 4, 5, 4, 5, 5, 6,
3, 4, 4, 5, 4, 5, 5, 6, 4, 5, 5, 6, 5, 6, 6, 7,
3, 4, 4, 5, 4, 5, 5, 6, 4, 5, 5, 6, 5, 6, 6, 7,
4, 5, 5, 6, 5, 6, 6, 7, 5, 6, 6, 7, 6, 7, 7, 8};
```

Example:
Integer Number: 2511
Binary Representation in 32 bit number.

00000000 00000000
00001001 11001111

As per this solution
6+2+0+0= 08

```
unsigned int num;
unsigned int ctr;    // Number of bits set.

ctr = LookupTable[num & 0xff] +
      LookupTable[(num >> 8) & 0xff] +
      LookupTable[(num >> 16) & 0xff] +
      LookupTable[(num >> 24)];
```



Problems Based on Arrays

❖ Problem: **Matching Nuts & Bolts Problem (Lock & Key problem) [3]-**

- Nuts represented as array of character
`char nuts[] = {'@', '#', '$', '%', '^', '&', '*', '+'}`
- Bolts represented as array of character
`char bolts[] = {'$', '%', '+', '&', '*', '^', '@', '#'}`



References

1. Seymour Lipschutz, “Data Structures”, Schaum’s Series McGraw Hill edition 2013.
2. www.cracktheinterview.com
3. <http://www.crazyforcode.com/matching-nuts-bolts-problem-lock-key-problem/>